

Movie Rental Design

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Section 1: Introduction:

The purpose of this project is to transition from an Operational Transaction Processing (OLTP) system to an Analytical Data Warehouse using Online Analytical Processing (OLAP) principles. The existing OLTP system is highly normalized to optimize daily transactional workflows, such as processing film rentals and capturing payments. However, this structure is inefficient for complex analytical queries.

This design introduces a Dimensional Model tailored for the movie rental business. By denormalizing the operational tables into a core Star Schema, we enable business managers to perform fast reporting, discover insights, and make data-driven decisions regarding film popularity, customer behaviors, and store revenues.

Section 2: Business Questions:

To support business strategic planning, the proposed data warehouse is designed to answer the following key business questions:

1. What are the top-performing film categories in terms of total rental revenue?

- *Importance:* Helps inventory managers understand user preferences and allocate budget towards buying more licenses for high-demand film categories.

2. Who are the top 10% highest-paying customers, and what is their rental frequency?

- *Importance:* Enables the marketing department to identify loyal customers and design targeted promotional campaigns or loyalty programs.

3. Which stores are generating the highest rental counts and payment revenues on a monthly basis?

- *Importance:* Assists executives in evaluating individual store performance, managing staff resources, and planning expansions or closures.

4. How do rental trends fluctuate across different seasons or days of the week?

- *Importance:* Allows store managers to optimize operations, predict peak rental days, and plan staffing effectively.

Section 3: Dimensional Model Design:

3.1 Selected Business Processes

The primary business process selected for this dimensional model is the **Movie Rental and Payment Event**, which captures the core revenue-generating mechanism of the business.

3.2 Fact Tables

1. Fact_Rental_Payment

- **Business Process:** Tracking both movie rental occurrences and their financial transactions.
- **Grain:** One row per individual rental payment transaction.
- **Measures:**
 - payment_amount (Additive - Total cash received)
 - rental_duration_days (Additive - Difference between return date and rental date)
 - rental_count (Degenerate/Additive - Constant value of 1 to easily sum total rentals)
- **Foreign Keys (Foreign Keys linking to dimensions):**
 - date_key
 - customer_key
 - film_key

- store_key

3.3 Dimension Tables

1. Dim_Date

- **Attributes:** date_key (PK), full_date, day_of_week, day_name, month, month_name, quarter, year.
- **Source OLTP Tables:** Created using generated calendar records alongside timestamps from rental and payment tables.
- **Purpose:** Allows time-series slicing (by year, quarter, month, or day) to analyze seasonal trends.

2. Dim_Customer

- **Attributes:** customer_key (PK), customer_id (Source key), first_name, last_name, full_name, email, active_status, city, country.
- **Source OLTP Tables:** customer, address, city, country.
- **Purpose:** Flattens customer profile and location info to understand customer demographics and segments.

3. Dim_Film

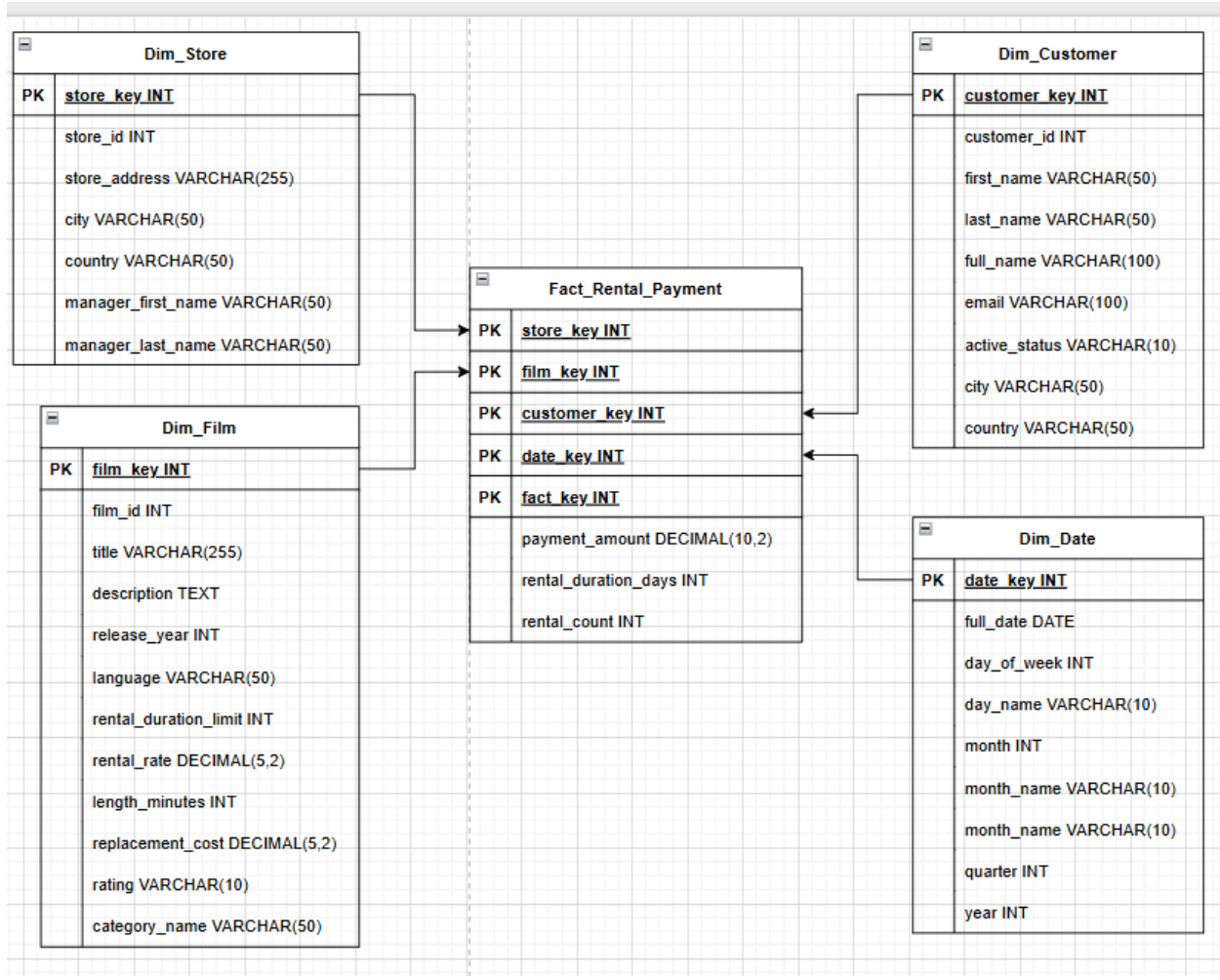
- **Attributes:** film_key (PK), film_id (Source key), title, description, release_year, language, rental_duration_limit, rental_rate, length_minutes, replacement_cost, rating, category_name.

- **Source OLTP Tables:** film, language, film_category, category.
- **Purpose:** Consolidates film information and genre groupings to track which types of movies perform best.

4. Dim_Store

- **Attributes:** store_key (PK), store_id (Source key), store_address, city, country, manager_first_name, manager_last_name.
- **Source OLTP Tables:** store, staff, address, city, country.
- **Purpose:** Denormalizes store locations and managers to evaluate geographical and organizational performance.

Section 4: Dimensional Model Diagram (Star Schema Description):



Section 5: ETL Design:

5.1 Extract

Data will be incrementally extracted from the source operational OLTP MySQL database. The extraction process targets the delta changes using transaction timestamps from the core tables: rental, payment, customer, film, inventory, and store.

5.2 Transform

- **Data Cleansing:** Handling missing values in return_date fields (where a movie hasn't been returned yet) by replacing nulls with a default placeholder or standard late indicator.
- **Denormalization (Merging):** Joining relational tables (e.g., combining customer, address, city, and country) into a single flat structure for dimensions.
- **Surrogate Key Generation:** Creating auto-incrementing integer values (e.g., film_key) to serve as Primary Keys in our data warehouse, isolating it from changes in operational source IDs.
- **Calculated Attributes:** Merging first_name and last_name into a unified full_name attribute for ease of end-user reporting.

5.3 Load

- **Phase 1 (Dimensions):** Load the dimension tables (Dim_Date, Dim_Customer, Dim_Film, Dim_Store) first. This ensures all dimension lookups are ready.
- **Phase 2 (Facts):** Query the newly generated surrogate keys from the dimension tables based on natural source IDs, and then insert records directly into **Fact_Rental_Payment**.

Section 6: Data Quality Considerations:

To maintain integrity in the data warehouse, the ETL process will implement the following strict data quality constraints:

1. **Completeness Check:** Reject or flag any incoming transaction if essential business foreign keys (such as customer_id or film_id) are missing or null.
2. **Referential Integrity Validation:** Ensure that every transaction row successfully resolves to an existing surrogate key in the dimension tables before allowing entry to the fact table.
3. **Logical Date Constraint:** Verify that return_date is chronologically equal to or greater than the rental_date.

Any anomalous backwards dates are pushed to an error logging table.

4. **Duplicate Safeguard:** Enforce unique constraints combination checks over the source operational transactions to prevent duplicate payment ingestion.

Section 7: Conclusion:

The proposed High-Level Data Warehouse structure transforms complex relational operational rows into an optimized, intuitive Star Schema design. By isolating analytical workloads onto this target model, business executives can seamlessly evaluate product success, track accurate store performance, and understand user behaviors without adding overhead to operational systems. Ultimately, this implementation equips the business with reliable, fast analytical capabilities to make confident data-driven investments.